VINCENT MANNARINO

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EDUCATION

McMaster University — Hamilton, ON **Sept. 2026 – Expected 2030**
Bachelor of Engineering (B.Eng.), Pursuing Mechatronics

Secondary School DiplomaJune 2026

- Ontario Scholar
- Honour Roll with Distinction (Gr. 11–12)
- Top 6 U/M Course Average: 97%
- AP Calculus AB Exam: Achieved a score of 5 (Highest Possible Score)
- DELF B1 Certified (French Immersion Graduate)
- Canadian Senior Mathematics Competition (CSMC): Certificate of Distinction (Top 25% Worldwide)

TECHNICAL SKILLS

Languages: Python, C++ , MicroPython, JavaScript, HTML/CSS
Software & Tools: Git/GitHub, SolidWorks (CAD), PyTorch, OpenCV, MediaPipe, scikit-learn, Flask, Arduino
Focus Areas: Machine Learning, Computer Vision, Embedded Systems, Circuit Design, Sensor Integration

PROJECTS

AI-Based Fall Detection System | Python, PyTorch, OpenCV, MediaPipe2026

- Engineered a real-time computer-vision pipeline using MediaPipe pose landmarks and a custom-trained PyTorch neural network to detect falls from webcam video, reaching 95.24% test accuracy while running entirely on a CPU
- Automated emergency email alerts on sustained fall detection, removing the need for a wearable device
- Awarded Best of Fair, 1st Place Senior Division, and Computer Science Award (2026 Algoma STEM Fair); selected to represent the region at the Canada-Wide Science Fair

AI Stock Predictor | Python, PyTorch, Flask2026

- Built a recurrent neural network (LSTM) from scratch in PyTorch, without relying on built-in layers, to predict next-day stock closing prices from historical OHLCV data
- Backtested predictions against a buy-and-hold baseline and deployed the model through a Flask web application for live ticker predictions

Automated Solar Tracking System | Arduino, C++, Circuit Design2024

- Designed and programmed an Arduino-based system that orients a solar panel toward the strongest light source using photoresistor sensors
- Applied the scientific method to quantify efficiency gains over a fixed-panel baseline; awarded 1st Place (Intermediate Division), Engineering Award, and Computer Science Award

LEADERSHIP & ACTIVITIES

VEX Robotics Club — Founding Member & Autonomous Programmer **2024 – 2026**

- Co-founded the school's VEX Robotics team; programmed and debugged autonomous match routines and contributed design improvements to the competition robot
- Won Tournament Champion and Autonomous Award titles across two regional competitions

Mayor's Youth Advisory Council (MYAC) — Elected Member **2024 – 2026**

- Represented youth perspectives in city council decisions and helped allocate community grants and scholarships

Operation Smile — Volunteer2025

- Helped organize a school-wide fundraising campaign supporting corrective surgeries for children with cleft lips and palates